

Synthesis and Crystal Structure of Dicopper(II) Compound Containing Oxalate and Reduced Imino Nitroxide Radicals^①

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ABSTRACT A new dicopper(II) compound with oxalate and reduced imino nitroxide radicals $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2](\text{NO}_3)_2$ (Him2-py=1-hydroxy-2-(2'-pyridyl)-4,4,5,5-tetramethylimidazoline) has been synthesized and its crystal structure was determined by X-ray diffraction method. The complex crystallizes in monoclinic, space group $P2_1/n$ with $a = 9.913(4)$, $b = 19.057(8)$, $c = 10.169(4)$ Å, $\beta = 114.292(7)^\circ$, $V = 1751(1)$ Å³, $\text{C}_{26}\text{H}_{38}\text{Cu}_2\text{N}_8\text{O}_{14}$, $M_r = 813.72$, $D_c = 1.543$ g/cm³, $\mu(\text{MoK}\alpha) = 1.290$ mm⁻¹, $F(000) = 840$, $Z = 2$, the final $R = 0.0398$ and $wR = 0.0986$ for 1878 observed reflections ($I > 2\sigma(I)$). The imino nitroxide 2-(2'-pyridyl)-4,4,5,5-tetramethylimidazoline-1-oxyl (im2-py) is reduced in the reaction to yield Him2-py which consists of oxalato-bridged centrosymmetric $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2]^{2+}$ ions and noncoordinated NO_3^- anions. Each copper(II) ion is in a distorted tetragonal pyramid environment: two nitrogen atoms from Him2-py and two oxygen atoms from the oxalate group are in the basal plane, and one water molecule at the axial position.

Keywords: crystal structure, reduced nitroxide radical, dicopper compound, oxalate bridge

1 INTRODUCTION

There has been increasing interest in molecular-based magnetic materials, in which the combination of metal ions and organic radicals are used to construct assembled systems^[1, 2]. Nitroxide radicals are normally used as spin carriers due to their exceptional stability and ease of chemical modification. Heterospin systems consisting of nitroxide radicals and paramagnetic metals ions constitute one of the most promising design strategies for high Curie-temperature molecularbased magnets^[3, 4]. However, nitroxide radicals can undergo redox

reactions with transition metal ions under certain experimental conditions^[5]. In this paper, we report an oxalato-bridged dicopper(II) compound, $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2](\text{NO}_3)_2$ with reduced nitroxide radicals.

2 EXPERIMENTAL

2.1 Materials and instrument

All reagents were of AR grade and used without further purification. $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ and 2-(2'-pyridyl)-4,4,5,5-tetramethylimidazoline-1-oxyl(im2-py) were prepared following literature methods^[6, 7].

Received 31 March 2003; accepted 12 July 2003

① This work was supported by the National Natural Science Foundation of China (No. 20271029)

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2.2 Synthesis of

$[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2](\text{NO}_3)_2$

An aqueous solution (10 mL) of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ (0.0459 g, 0.1 mmol) was added to a mixture of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ (0.072 g, 0.3 mmol) and im2-py (0.065 g, 0.03 mmol) in 15 mL of methanol. The solution was stirred for 1.5 h and filtered. The brown filtrate was kept at room temperature for two weeks to yield well-formed dark green crystals and the solution turned green. Anal. Calcd. (%) for $\text{C}_{26}\text{H}_{38}\text{Cu}_2\text{N}_8\text{O}_{14}$: C, 38.37; H, 4.71; N, 13.76. Found (%): C, 37.75; H, 4.79; N, 13.94.

2.3 Crystal structure determination

A dark green crystal ($0.25\text{mm} \times 0.20\text{mm} \times 0.20\text{mm}$) was selected and mounted on a glass fiber. All measurements were made on a Bruker Smart 1000 diffractometer equipped with a graphite-monochromated $\text{MoK}\alpha$ radiation ($\lambda = 0.71073 \text{ \AA}$). A total of 5925 reflections were collected in the $2.14 < \theta < 23.36^\circ$ range at room temperature and 2528 were independent ($R_{\text{int}} = 0.0383$), of which 1878 were observed ($I > 2\sigma(I)$). The absorption correction was performed empirically. The structure was solved by direct methods with SHELXS-97 program^[8] and refined with SHELXL 97^[9] by full-matrix least-squares technique on F^2 . All non-hydrogen atoms were refined anisotropically, while the hydrogen atoms were located geometrically and refined isotropically. The final $R = 0.0398$ and $wR = 0.0986$ ($I > 2\sigma(I)$) with $R = \sum(|F_o| - |F_c|)/\sum|F_o|$ and $wR = (\sum w(|F_o|^2 - |F_c|^2)^2 / \sum w|F_o|^2)^{1/2}$, $w = 1/[\sigma^2(F_o^2) + (0.0616P)^2 + 2.4020P]$ where $P = (F_o^2 + 2F_c^2)/3$. $(\Delta/\sigma)_{\text{max}} = 0.002$, $S = 0.921$, $(\Delta\rho)_{\text{max}} = 0.391$ and $(\Delta\rho)_{\text{min}} = -0.280 \text{ e/\AA}^3$. The positional parameters

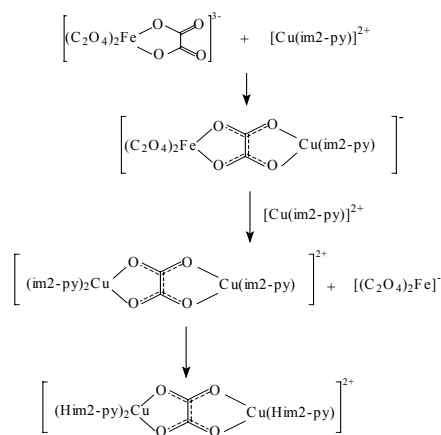
and selected important bond parameters are given in Tables 1 and 2, respectively.

3 RESULTS AND DISCUSSION

3.1 Synthesis of

$[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2](\text{NO}_3)_2$

The formation of dicopper(II) compound most likely arises from the electrophilic attack of $\text{Cu}(\text{im2-py})_2^{2+}$ complex on the coordinated oxalate ion in $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]^{10}$. The reasonable mechanism is given in Scheme 1. In the meantime, the imino nitroxide of the compound was reduced to Him2-py. A similar phenomenon has been observed in some metal complexes with nitroxide radicals^[11~13]. The mechanistic details of the reduction of nitroxide radical are not completely clear, but the reduction of nitroxide radical has been shown to be promoted by acidic species^[14]. It is likely that the formation of Him2-py, in the present case, is favored by acidic impurities and standing for a long time.



Scheme 1

Table 1. Atomic Coordinates ($\times 10^4$) and Equivalent Isotropic Displacement Parameters ($\text{\AA}^2 \times 10^3$)

Atom	x	y	z	U_{eq}	Atom	x	y	z	U_{eq}
Cu(1)	3095(1)	4118(1)	8295(1)	35(1)	C(6)	-115(6)	3840(3)	3122(7)	78(2)
O(1)	4368(3)	4212(1)	10401(3)	36(1)	C(7)	2005(9)	3469(4)	2545(6)	99(2)
O(2)	3938(3)	5037(1)	8146(3)	40(1)	C(8)	1947(4)	3203(2)	5996(4)	33(1)
O(3)	708(4)	2429(2)	4031(3)	58(1)	C(9)	1932(4)	2771(2)	7191(5)	34(1)
O(4)	1241(4)	4556(2)	8731(5)	66(1)	C(10)	1452(5)	2091(2)	7119(5)	46(1)
N(1)	2401(4)	3847(2)	6251(4)	36(1)	C(11)	1548(6)	1767(3)	8377(6)	56(1)
N(2)	1685(4)	2983(2)	4641(4)	44(1)	C(12)	2126(6)	2128(3)	9652(6)	55(1)
N(3)	2491(4)	3129(2)	8458(4)	34(1)	C(13)	2587(5)	2813(2)	9651(5)	46(1)
C(1)	4874(5)	5238(2)	9352(4)	32(1)	N(4)	2745(6)	1036(3)	3810(5)	71(1)

C(2)	2523(5)	4137(2)	4943(5)	41(1)	O(5)	1824(6)	1041(3)	4302(6)	115(2)
C(3)	2051(7)	4897(3)	4673(6)	67(2)	O(6)	2969(9)	1532(3)	3227(7)	165(3)
C(4)	4159(6)	4076(3)	5227(7)	74(2)	O(7)	3443(6)	489(3)	3915(6)	115(2)
C(5)	1519(6)	3617(3)	3734(5)	49(1)					

U_{eq} is defined as one third of the trace of the orthogonalized U_{ij} tensor.

Table 2. Selected Bond Lengths (Å) and Bond Angles (°)

Bond	Dist.	Bond	Dist.	Bond	Dist.
Cu(1)–N(1)	1.971(3)	Cu(1)–O(4)	2.224(4)	Cu(1)–N(3)	2.003(3)
Cu(1)–O(1)	1.997(3)	Cu(1)–O(2)	1.974(3)	O(3)–N(2)	1.394(5)
Angle	(°)	Angle	(°)	Angle	(°)
N(1)–Cu(1)–O(2)	97.28(13)	O(1)–Cu(1)–O(4)	87.81(14)	O(2)–Cu(1)–O(4)	95.30(15)
O(2)–Cu(1)–O(1)	84.35(11)	N(1)–Cu(1)–O(1)	161.31(14)	N(3)–Cu(1)–O(4)	92.11(15)
O(2)–Cu(1)–N(3)	172.40(14)	N(1)–Cu(1)–N(3)	81.65(14)		
N(1)–Cu(1)–O(4)	110.49(16)	O(1)–Cu(1)–N(3)	94.27(12)		

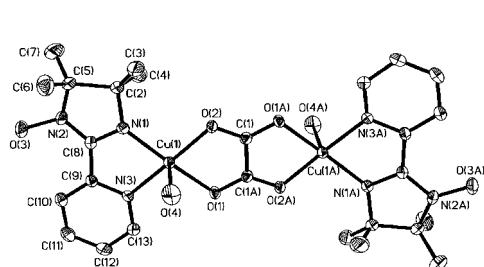


Fig. 1. The labeled diagram of the dinuclear unit $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2]^{2+}$ with the thermal ellipsoids are at 30% probability

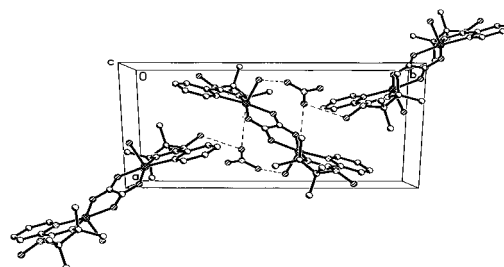


Fig. 2. A sketch of the unit cell of compound $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2](\text{NO}_3)_2$

3.2 Crystal structure of $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2](\text{NO}_3)_2$

The perspective view of $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2]^{2+}$ with the labeling scheme is shown in Fig. 1 with two noncoordinated nitrate anions being omitted. The structure consists of centrosymmetric $[\text{Cu}_2(\mu\text{-C}_2\text{O}_4)(\text{Him2-py})_2(\text{H}_2\text{O})_2]^{2+}$ ions and noncoordinated NO_3^- anions. Each Cu(II) ion is in 4+1 environment with the O(1), O(2), N(1) and N(3) atoms in the basal plane and O(4) from water molecule at the axial position. The in-plane Cu–O bond distances are 1.997(3) and 1.974(3) Å for Cu(1)–O(1) and Cu(1)–O(2), respectively, and Cu–N bond lengths are 1.971(3) and 2.003(3) Å for Cu(1)–N(1) and Cu(1)–N(3), respectively. The axial Cu–O bond distance is 2.224(4) Å. Thus, the coordination geometry around the copper(II) ion can be described as a distorted tetragonal pyramid. The

N–O bond distance is 1.394(5) Å, indicating clearly the reduced form of the radical; while the corresponding distance in nitroxide ranges from 1.25 to 1.32 Å^[15, 16]. The reduced radical can exist in two tautomeric forms^[14], the amidino-oxide and imino-hydroxylamine. Since the nitrogen atom from the imino group here is involved in the coordinating copper(II) ion, the reduced radical form should be the latter. The copper-copper separation is 5.179 Å.

A sketch of the unit cell is shown in Fig. 2. The reduced radicals form intramolecular hydrogen bonds with the oxygen atoms from nitrate groups (2.837 Å for O(3)⋯O(5)), and the coordinated aqua molecules form intermolecular hydrogen bonds with the oxygen atoms from nitrate groups of the adjacent units (2.857 Å for O(4)⋯O(7) ($x-1/2, -y+1/2, z+1/2$) and 2.894 Å for O(4)⋯O(7) ($-x+1/2, y+1/2, -z+3/2$)).

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